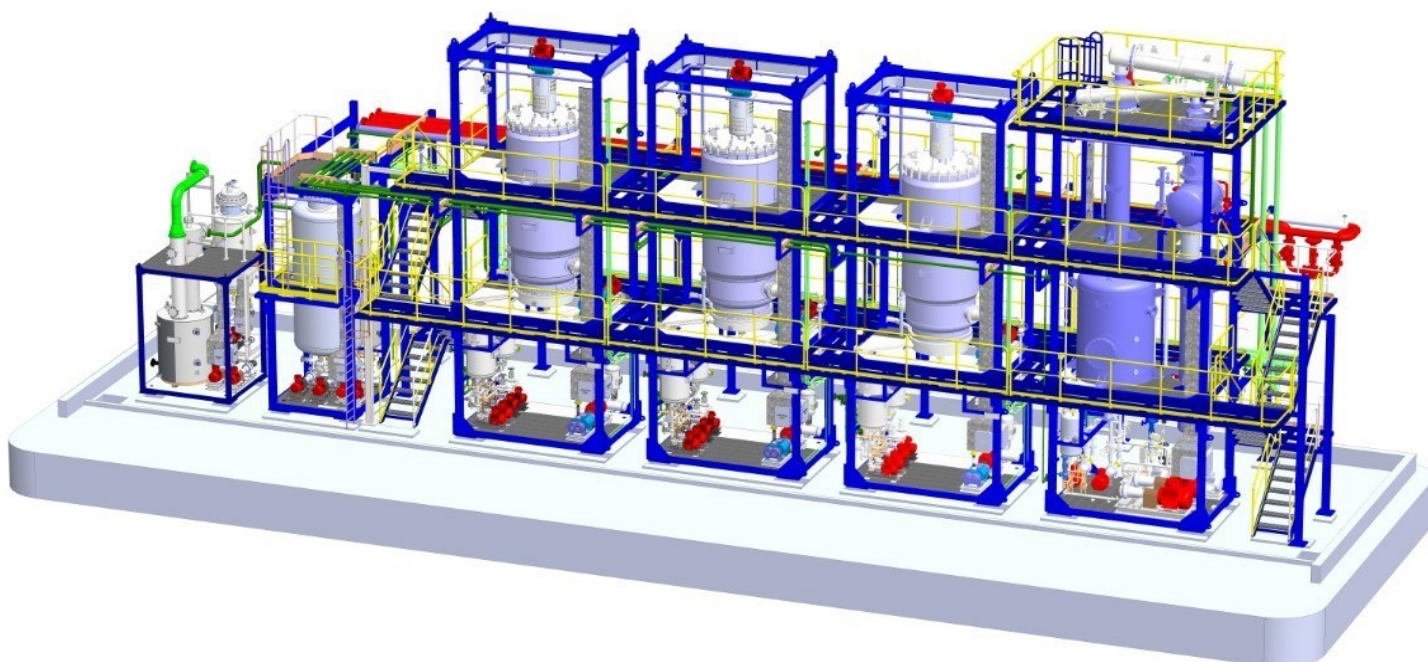


Sub.: Thermopac Technology For Used Oil Re-Refining Project Budgetary Offer For
Engine Oil Refinery PLC SCADA Controlled Automatic Continuous

We are pleased to submit our offer for design, manufacturing, supply and commissioning of used oil re-refining plant to be installed and commissioned.

This technology is developed to process used engine oil to its maximum output by means of environmentally friendly fully automatic continuous process. Our technology uses specially designed **TWFE (Thermopac Wiped Film Evaporator) process** to optimize the yield.

The process and its advantages along with all technical details are described in the following pages.



For **THERMOPAC**
(Turnkey Engineering Solution Division)

Marketing Manager

Raw Material Required: -

The oil refining plant uses used lubricating oil as a raw material. The used oil can typically come from garages, service stations, mechanic shops, and industrial users. The used oil typically contains water, diluents, solid particles from rust and corrosion, dirt and other suspended and dissolved impurities.

Typical Analysis of Used Lube Oil:

Appearance	:	Viscous liquid with suspended impurities
Color	:	Black
Specific gravity (D-1298)	:	0.850 to 0.900
Water content (% in emulsion w/w)(D-4006)	:	5 to 7%
Flash Point, °C (D-92)	:	100 to 190
Viscosity, cSt at 40°C	:	70 to 110
At 100 °C (D-445)	:	9.5 TO 12.5
Ash sulphated, % w/w (D-482)	:	1.5 to 3.0
Pentane insoluble, % w/w (D-893)	:	1.0
Total acid no. mg KOH/g	:	1.0

The Design Specifications:

The plant is designed based on typical feed characteristics as above and for input specification given below.

Input: used oil of mixed hydrocarbons containing:

- ~ 5 % water
- ~ 5 - 8% light ends [gasoline, aromatics]
- ~ 5 - 8% diesel, kerosene
- ~ 70 - 80% long-chain hydrocarbons
- ~ 5 - 10% residual solids and asphaltenes.

Output specifications:

From Skid -1:

- ~ 5% water containing traces [< 2%] of hydrocarbons;
- ~ 5% - 10 % lights [naphthalene, gasoline, kerosene, and diesel]

From Skid - 2:

- ~ 50% - 55% medium chain hydrocarbons (neutral 150)
- ~ 50% - 45% heavy used oil [containing heavy base oil, solids and asphaltenes] OR
- ~ 30% - 35% longer chain hydrocarbons (neutral 300)
- ~ 10% - 15% asphalt extender [containing solids and asphaltenes]

Process:

Our Plant processes used lube oil, using high vacuum molecular distillation technology in the following six stages:

Stage 1: De-hydration and De-gas oil Skid I

Stage 2: Distillation Skid-II

Stage 3: NMP Solvent Extraction

Stage 4: Finishing for Group-II +

Stage 5: Pollution Skid -IV Scrubber and Thermal Oxidizer

This process technology re-refines used lube oil to its maximum output by means of environmentally friendly PLC controlled fully automatic, continuous process.

The plant processes used lube oil and industrial oils, using high vacuum molecular distillation technology. This process is based on three different modular skids for three important stages of process:

Item class A: Standard scope supply

Excluding site fabrication, piping, electrical and civil jobs

Stage 1: Skid - I: De-hydration and De-gas Oil Skid. (Front End Skid)

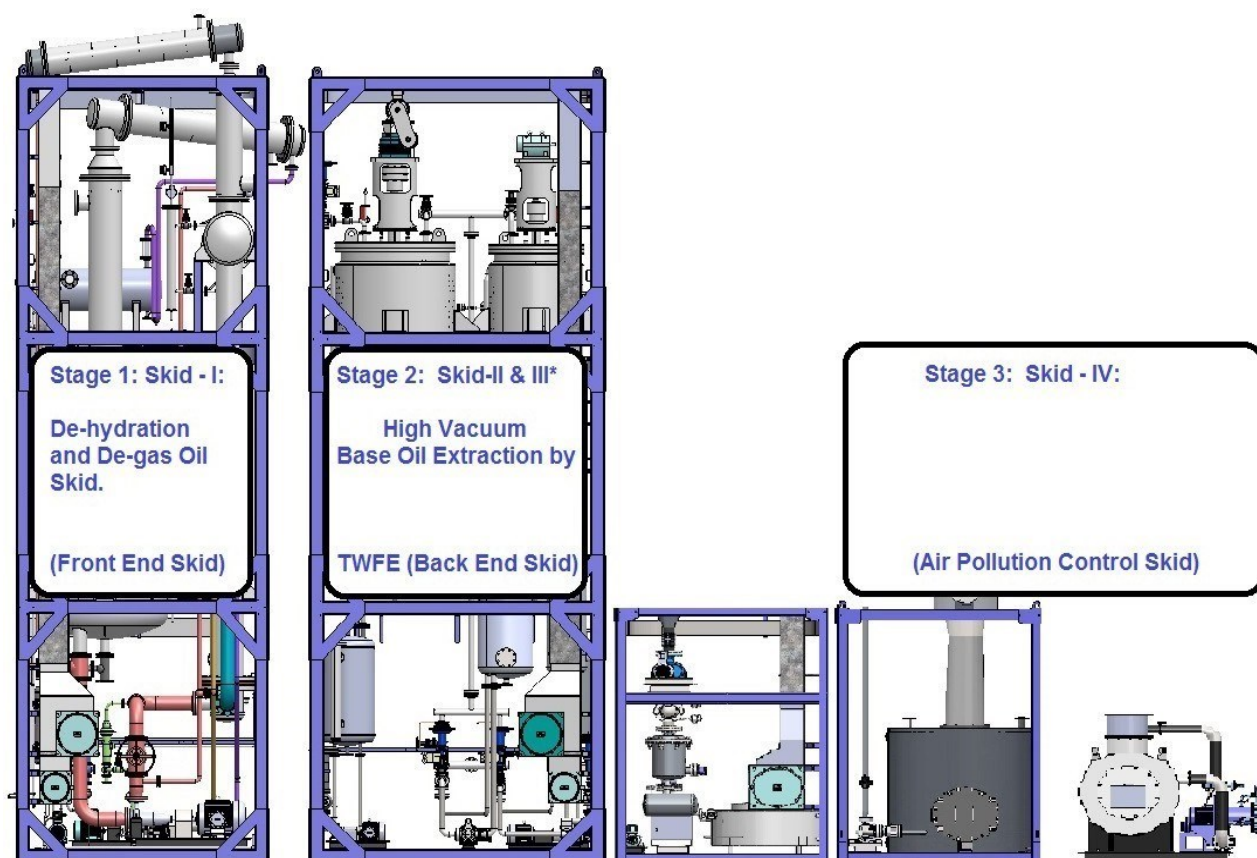
Removes traces of water, naphtha, diesel oil and gas oil by flash evaporation; and separates water: and diesel, naphtha and gas oil as different distillation cuts and sends all uncondensed gases to pollution skid .

Stage 2: Skid-II) Base Oil Extraction by TWFE (Back End Skid) Dehydrated and free from water, naphthalene and gas oil, the used lube oil will be heated to a specific temperature and sent to a specially designed evaporator to extract light lube oil. This cut is named as processed light lube oil and sent to storage tank for subsequent polishing system to generate Light Base Oil.

Now the remaining used oil containing heavy lube oil and residue will be sent to specially designed Thermo Wiped Film Evaporator, where the heavier hydrocarbon fractions will evaporate and condense and collect into a receiver. This cut is named as processed heavy lube oil and sent to storage tank for further polishing system to generate Heavy Base Oil.

Bottoms of this Thermo Wiped Film Evaporator is basically residue which will be transferred to asphalt tank for further processing into RFO.

Vacuum system on this skid can maintain desired vacuum and temperatures; all exhaust from this vacuum system is sent to the pollution skid



NMP SOLVENT EXTRACTION PLANT

For Processed Used Oil – High Quality Base Oil Production

NMP (N-Methyl-2-Pyrrolidone) solvent extraction selectively removes aromatic compounds, PAHs, sulphur, nitrogen, colour bodies and other polar impurities from vacuum distilled used oil to produce high quality base oil with improved viscosity index, lower sulphur and excellent oxidation stability.



CLEANER OIL
Removal of aromatics, PAHs, sulphur & polar compounds



BETTER PERFORMANCE
Improved Viscosity Index, oxidation stability & thermal stability



HIGH YIELD
Maximum base oil recovery with low solvent loss



RELIABLE & SAFE
Closed system operation with advanced control & safety features

FEED (VACUUM GAS OIL – VGO)

ASTM Colour	4.5 – 5.5
Viscosity @ 40°C	32 – 42 cSt
Viscosity Index (VI)	115
Sulphur	4000 PPM

OUTPUT (EXTRACTED BASE OIL)

ASTM Colour	≤ 1.0
Viscosity Index (VI)	125
Sulphur	1500 PPM
Appearance	Bright & Clear

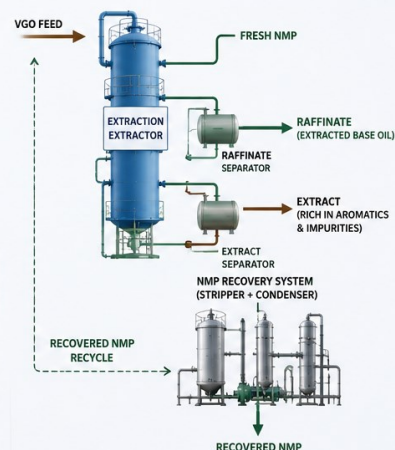
KEY BENEFITS

- Significant reduction in colour, aromatics & sulphur
- Improved VI and oxidation stability
- High quality base oil suitable for further finishing or blending
- Low NMP consumption with high recovery
- Environmentally responsible closed-loop system

SOLVENT EXTRACTION – SINGLE EXTRACTOR SYSTEM



TYPICAL PROCESS FLOW DIAGRAM



CAPACITY
5,000 LPH
VGO Feed



SOLVENT
NMP
(N-Methyl-2-Pyrrolidone)

Cleaner Oil • Better Performance • Higher Value • Sustainable Future



Driving Circular Economy
Through Cleaner Technology

Stage 3: NMP Solvent Extraction System for Processed Used Oil

The NMP (N-Methyl-2-Pyrrolidone) Solvent Extraction System is a liquid-liquid extraction process designed to upgrade vacuum-distilled used lubricating oil fractions into high-quality base oils. The process selectively removes aromatic hydrocarbons, polycyclic aromatic compounds (PAHs), sulphur- and nitrogen-containing compounds, colour bodies, oxidation products, and other polar impurities while retaining the desirable paraffinic base oil.

The extraction is carried out in liquid to liquid extraction, where the processed oil is contacted with NMP solvent. The purified oil (raffinate) is then separated from the solvent-rich extract phase. Both streams pass through solvent recovery systems, where NMP is recovered, purified, and recycled back to the extraction process with minimal solvent make-up.

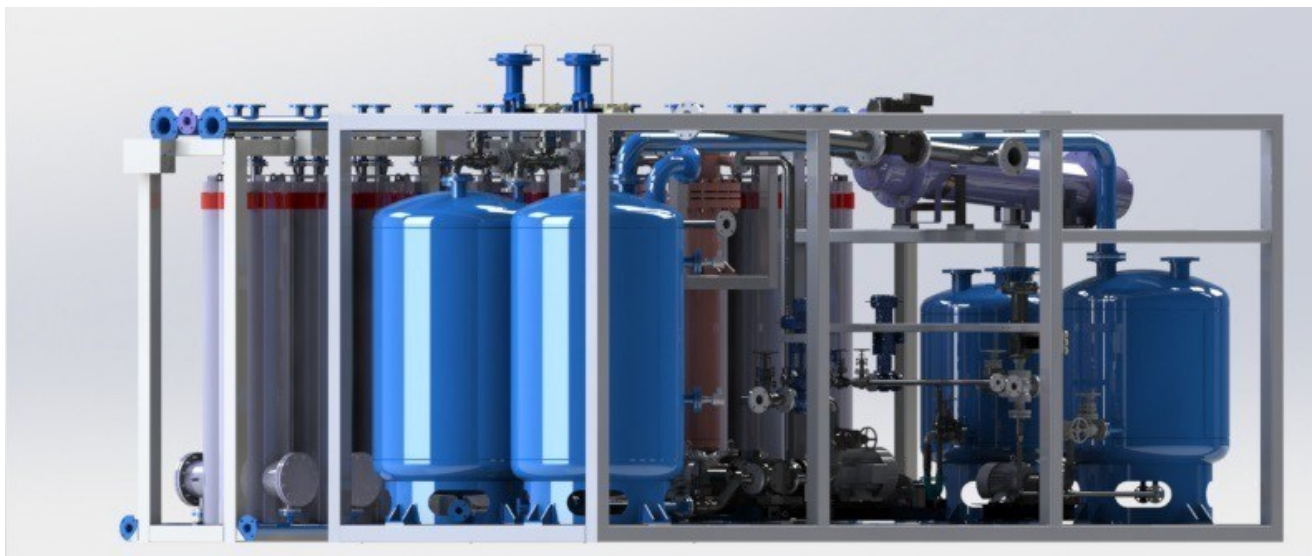
The system is designed to improve base oil quality by increasing viscosity index (VI), reducing colour, improving oxidation stability, and lowering aromatic and heteroatom content, resulting in premium-quality base oil suitable for further finishing or blending. The plant includes extraction columns, solvent recovery units, heat exchangers, pumps, instrumentation, automation, and all associated utilities required for continuous operation.

Stage 4: Fully Automatic Non Consumable Regenerateable Finishing System

Adds value to reclaimed oils by improving color and reducing or removing odors

The clay polishing systems provides an economical alternative to hydrogen treatment plants as a means of adding value to re-refined lube oil.

The system is based upon banks of active columns that contain adsorptive media. The physical characteristics of the media allow it to be re-activated once saturated, and thereby permitting several hundred cycles to be run before being replaced.



Once exhausted, typically of 6 months to a year of continuous operation, the media is disposed off in a conventional landfill site, as a dry non-hazardous waste.

Processing oil through the system results in oil that is stable against oxidation over any measured timescale - this increase in shelf life is an added benefit of the polishing system. Furthermore, the Viscosity Index is not degraded by the polishing system.

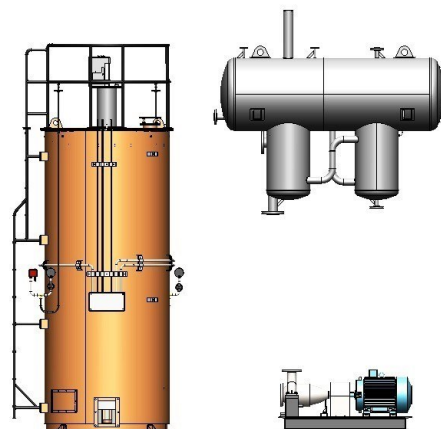
Stage 5: Skid - IV: (Pollution Skid)

All vacuum exhaust and dehydration skid vent gases will be scrubbed to absorb SO_x and H₂S compounds through specially designed scrubbing system and balance exhaust gas will now contain non condensable VOC. This will be sent to thermal oxidation.

High Temperature Thermal Oil System

The Thermal Oil Heating system consists of the following equipment:

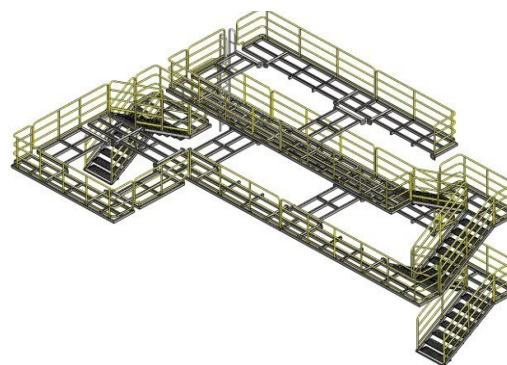
1. High Temp Heater
2. High Temp Circulation Pump
3. Expansion cum Deaerator tank



Supply of Thermal Oil fluid is excluded from our standard supply.

Platform, Walkway, Ladder and Equipment Lifting arrangement around refinery Fixed platforms, walkways, stairways and ladders, complete with:

- Guard railing
- Fall protection from ladders
- Ladder cages and enclosures
- Access hatches



Supply of chain hoist is excluded from our standard supply.

Parallel or Series operations can be achieved through the system as it is designed to switch from parallel to series mode operation via the custom-built SCADA supplied with every system. This allows the plant owner to easily switch between different processing modes for each oil stream reclaimed through the front-end system. Integration with existing front-end systems and/or tank farms is easily achieved.

The out product characteristics will completely depend on the feed properties.

The viscosities of the **Light Base Oil** and the **Heavy Base Oil** can be adjusted to desired values by controlling the temperatures and vacuums.

The viscosities of the **Light Base Oil** and the **Heavy Base Oil** fully depend on input feed oil (used oil feed)

We can produce Heavy Oil as heavy as 80 -120 cSt at 40 deg C with flash points ranging from 210 - 240 deg C. However, the output percentage fully depends on long-chain hydrocarbon content of the feed. System is designed to handle 15% free and emulsified water in lube oil along with light ends such as diesel, naphtha and gas oil.

Control and Automation System

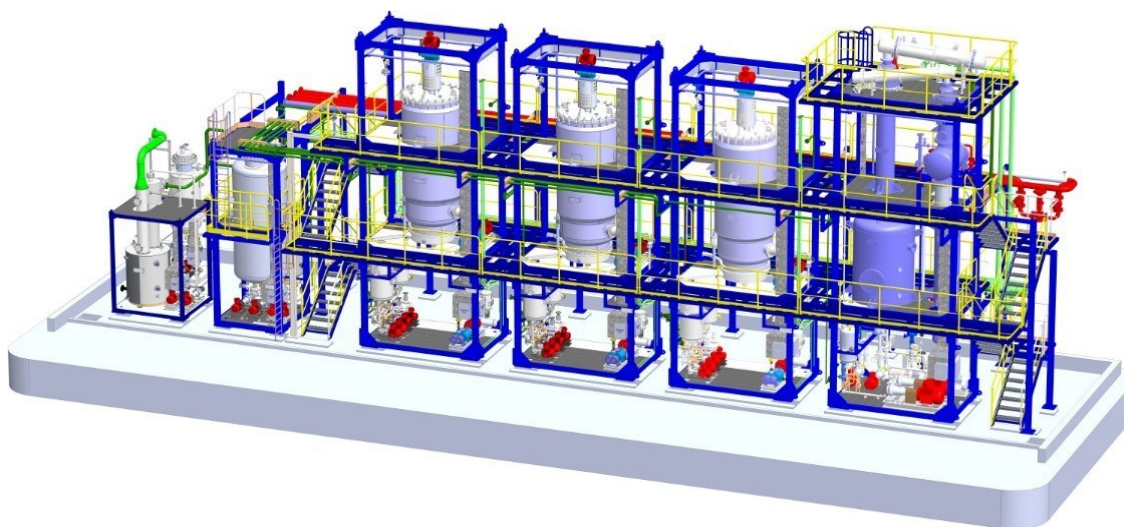
The operation of the Plant is fully automatic and remote controlled by a PLC and Scada, which centralizes all control and monitoring systems to allow Plant operation in a steady, reliable and safe manner.

Air compressors, Thermal Oil Heater, Thermal Oxidizer and Cooling tower will be controlled by PLC and Scada.

Controller configuration memory will have battery backup so that the controller maintains its configuration, state and information in the event of power failure.

OUR SKID MOUNTED UNITS COMPRISE:

All skids are pre-wired and are complete with necessary flame proof instruments and controllers such as temperature, pressure, flow and level and local junction boxes. All hot surfaces are insulated and clad wherever required. Input and output piping along with control and one isolation valve wherever required is part of our supply.



External refinery piping shown in above model is excluded from our standard supply,

CONVERTING USED OIL INTO PREMIUM GROUP II BASE OIL

NMP SOLVENT EXTRACTION + CPS POLISHING TECHNOLOGY

**USED OIL / VGO
(Before Treatment)**



- ASTM Colour 4.5 ~ 5.5
- Viscosity @ 40°C 32 ~ 42 cSt
- Viscosity Index (VI) 115
- Sulphur 4000 ppm

DARK COLOUR | HIGH SULPHUR | HIGH AROMATICS
Lower VI, Poor Oxidation Stability

**NMP EXTRACTION
+
CPS POLISHING**

**Cleaner Oil
Better Performance
Higher Value**

**GROUP II BASE OIL
(After NMP Extraction
& CPS Polishing)**



- ASTM Colour ≤ 1.0
- Viscosity @ 40°C 32 ~ 42 cSt
- Viscosity Index (VI) 125
- Sulphur < 200 ppm
- Appearance Bright & Clear

LOW COLOUR | LOW SULPHUR | LOW AROMATICS
Higher VI, Excellent Oxidation Stability



ADVANCED REFINING TECHNOLOGY
Removes aromatics, PAHs, sulphur and polar impurities for superior base oil quality.



IMPROVED PERFORMANCE
Higher Viscosity Index, better oxidation stability, and longer service life.



ENVIRONMENTALLY RESPONSIBLE
Closed-loop solvent system with >98% NMP recovery and very low losses.

FROM WASTE TO VALUE – PRODUCING PREMIUM GROUP II BASE OIL >>>

OUTPUT PRODUCT SPECIFICATIONS:

Typical specification of the output products

• Light Lube SN 150

	Refined base oil prior to finishing
Viscosity @ 40 deg cSt.	~25 - 37
Flash Point deg C – closed cup	>180°C
Density @ 15 deg C, kg/l	~ 0.85

• Heavy Lube SN 400-500

	Refined base oil prior to finishing
Viscosity @ 40 deg cSt	~80 -110
Flash Point – closed cup	>230°C
Density @ 15 deg C	~ 0.88

BY-PRODUCTS

The Plant will also produce the by-products with the typical characteristics as follows. The actual characteristics will be depending on the feedstock.

1. Light Fuel

Specific gravity	0.760 - 0.770
Flash point, °C	20

2. Gasoil

Specific gravity	0.840 - 0.860
Viscosity, cSt at 40°C	3 - 6
Flash point, °C	55 min
TAN, mg KOH/gr	0.1
LHV, kcal/kg	10,000

Asphaltic Residue

Specific gravity	0.900 - 1.050
Viscosity, cSt @100°C	300
Ash content, wt%	5 - 10
Sulfur, wt%	1 - 2
Softening point, °C	10 - 15
LHV, kcal/kg	8,500
Metals, wt%	2 - 4

OR

4. 180 CST Fuel Oil

Furnace oil is a dark viscous residual product used as a fuel In different types of combustion equipment. kinematic viscosity range of 125 to 180 CSt at 50°C

OUR SERVICES

- Documentation
 - Plot Plans and General Arrangement and PFD's
 - Material and Heat Balances
 - Equipment Data Sheets and Specification Sheets
 - P&I's and Piping Specs and Layout drawings
 - Package Units Specifications
 - Instrumentation Drawings
 - Electrical Single Line Diagrams, Electrical Drawings
 - Insulation & Painting Specifications
 - Operating Manuals
 - Mechanical Catalogues.
- Our Exact Scope in Construction, Erection, Pre-commissioning and Commissioning.
 - Process design and technical know-how
 - Mechanical design, engineering and structural basic design.
 - Electrical basic design
 - Instruments and control systems basic design.
 - Procurement and supply of the plants and equipments
 - Supervision of construction, erection/installation of the plant ▪ Startup / commissioning of the plant ▪ Operation / training to the staff.
- Pre-commissioning and Commissioning Assistance:
 - 4-week onsite assistance for final hookup of utilities and instruments and control wiring.
 - 2-week onsite assistance for startup
 - 2-week onsite assistance for demonstration and training.

MAJOR EMISSIONS & EFFLUENTS

- a) Emission from thermic fluid heater chimney.
- b) Very light ends storage tank and Skid vents pass through thermal oxidation system
- c) Distilled water recovered from the used oil containing traces (<2%) of hydrocarbons should be routed to a used water storage and/or treatment system (Out of our scope)

KEY EQUIPMENTS OF THE PROCESS

<u>Equipment Description</u>	<u>Country of Origin</u>
▪ Evaporator	Thermopac India
▪ Wiped Film Evaporator	Thermopac India
▪ Vacuum Pumps	Korea
▪ Vacuum Booster	USA
▪ Pumps	India
▪ High Temperature Pumps	KSB India
▪ Thermic Fluid Heater	Thermopac India
▪ Mechanical Seals	Eagle, India
▪ Switchgear	India
▪ Control Panel	Thermopac India

PROCESS GUARANTEES:

- Guaranteed throughput in the range of +/- 10%
- Asphalt viscosity @ 30°C, min > 1000 cSt
- Guaranteed 70% yield of base oil
- Maximum 15% yield of asphalt
- Fuel oil consumption, maximum 0.05 kg/kg ULO

CODES AND STANDARDS

- American Society of Testing Materials (ASTM)
- American Society of Mechanical Engineers (ASME)
- American Petroleum Institute (API)
- Tubular Exchanger Manufacturer's Association (TEMA)
- Local Standards where the plants will be located

ADVANTAGES OF TWFE PROCESS

- Sludge free & 100% environment friendly
- Highly efficient latest technology
- PLC automated, with SCADA system
- Skid mounted - Expandable, reduces set-up cost and time
- Modular construction
- Handles a variety of used lube oils (ULO)

SHIPMENT DETAILS

We will be shipping the entire project from our factory near Mumbai, India. Even when the price quoted is on Exworks basis (nearest port to your plant location), further transportation to your project site, loading, unloading, leading and placing of the equipment on foundations are to be arranged by the client at his cost.

SCHEDULE AND UNSCHEDULED MAINTENANCE

The plant should be taken up for cleaning / greasing/ oiling on schedule maintenance as follows:

Schedule	No. of days	Total days in year
Monthly	1	12
Quarterly	4	16
Unscheduled		10
Total down time days in a year		38 Days

SPARE PARTS:

Essential spare parts required for 1 year of trouble free operation of refinery and equipment's is part of supply

TIME SCHEDULE OF THE PROJECT

- 6 to 8 Months for shipment
- 1.5 Months shipping time
- 1 Month for Erection, piping, testing and commissioning.
- 1 Month contingency allowance

Total time Total time from issuing the P.O with advance to commissioning of plant *9.5 months

*The schedule as above is subject to force majeure as well as order backlog at the time of finalization of order.

WARRANTY:

All plant and equipment are guaranteed for performance as per specification provided in the offer and we will demonstrate the performance of the plant for 48 hours during the trial run.

Performance demonstration will

include:

- Throughput capacity
- Product and byproduct specification
- Consumption of utilities

All performance figures on capacity, product specifications, and utility consumption are subject to a tolerance of $\pm 10\%$

All components of the plant are covered under limited warranties given by our suppliers or by us All components and plant are guaranteed by THERMOPAC for period of 12 Months from date of commissioning or 18 months from date of shipment whichever is earlier.

Above warranty covers replacement or repair of components at free of cost and does not cover any liability towards any damage to personnel, property, or loss in production.